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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

B. Tech. Project

"AURA-3: AI-Powered Decentralized Autonomous Organization (DAO) for Startup Funding"

CA4 Progress Report - October 2026

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Project ID 24. CA4 submission. Report structure follows the supplied monthly progress report.

1. Work Distribution

The group continued the AURA-3 implementation during CA4, extending the earlier startup evaluation workflow with completed DAO implementation and a document extraction engine developed by the team. The distribution below continues the responsibilities recorded in the earlier report.

Member and PRN	Work distribution for CA4
Adesh Jain 23070126006	Continued the literature review on DAO funding and governance, consolidated requirements, and contributed to research documentation and the evaluation criteria.
Arav Saxena 23070126018	Led the AI architecture and startup evaluation workflow. Developed TitanOCR and utilised it for pitch-deck extraction, integrating document text and spatial evidence into the evaluation workflow.
Kartik Mehta 23070126056	Led the DAO governance and funding workflow, including proposal processing, voting and blockchain funding records. Continued database and governance integration work.
Mahi Laddha 23070126069	Continued the interface and presentation work, maintained project documentation, and consolidated the CA4 progress report and demonstration flow.

CA4 implementation summary

DAO implementation has been completed. TitanOCR has also been developed and utilised for pitch-deck evaluation. These additions connect document-based startup assessment with the governance workflow. The remaining work focuses on end-to-end validation, reproducible evaluation and final documentation.

Names, PRNs and original responsibilities are retained from the supplied report. Completion updates reflect the CA4 information supplied by the project team.

2. Introduction and Problem Statement

Introduction

AURA-3 is an AI-powered Decentralized Autonomous Organization for startup funding. It combines an off-chain startup evaluation engine with blockchain governance so that participants can examine a proposal before voting and inspect the resulting governance record. The project focuses on an academic prototype rather than a production investment service.

The earlier implementation established a frontend, FastAPI backend, retrieval workflow, startup analysis, investor reports and a chatbot. CA4 extends that work in two areas: completion of the DAO implementation and utilisation of TitanOCR, a standalone OCR and document-structure engine developed by the team, for reading pitch decks.

Pitch decks often combine selectable PDF text, screenshots, financial tables and dense multi-column slides. A plain text export can detach a number from its unit, lose a table header or obscure which slide supported an evaluation. TitanOCR addresses the extraction layer by retaining the location and origin of recognised content. The AI layer then interprets that evidence in the startup context.

Problem statement

Early-stage funding decisions can be difficult to inspect because the supporting documents, evaluator reasoning and final decision are scattered across different systems. Decentralized voting alone does not resolve this information gap. Participants also need readable evidence, visible uncertainty and a clear relationship between an assessment and the proposal under consideration.

The project therefore investigates how a document-grounded AI evaluation workflow can support a DAO funding process. The central challenge is to preserve the link between source material, analysis, governance decisions and funding records without treating model output as verified business truth.

CA4 scope

This report covers the implemented prototype, TitanOCR integration, research rationale and the validation work required to support final claims. It does not claim demonstrated investment returns, elimination of bias, an independent smart-contract audit or a production deployment.

3. Objectives and Research Questions

Objective	Expected project behaviour
Document-grounded evaluation	Extract pitch-deck evidence and use retrieval to support analysis of the business model, team, market, traction and risk.
Traceable document processing	Retain page numbers, bounding boxes, origin and confidence through the TitanOCR Document IR and its text projections.
DAO governance	Support proposal creation, token-based voting and funding records through the implemented DAO workflow.
Investor explanation	Present an evaluation report and retrieval-based answers so participants can inspect the basis of an assessment.
Controlled funding workflow	Connect governance decisions to treasury and milestone release rules, with review of off-chain evidence before execution.
Reproducible research	Evaluate extraction quality, evidence support and governance behaviour using explicit datasets and repeatable tests.

Research questions

RQ1. Does retaining spatial provenance improve the ability to locate and verify statements in a pitch-deck evaluation compared with plain extracted text?

RQ2. How reliably does the extraction workflow preserve numeric values, table associations and reading order across digital, scanned and hybrid pitch decks?

RQ3. Does presenting document-grounded evaluation alongside a proposal improve voter understanding and consistency in a controlled study?

RQ4. Do governance and milestone checks prevent invalid state transitions and repeated fund-release actions in the prototype?

These are research questions. The supplied material does not establish their answers. Claims about bias reduction, participation or investment performance require separate empirical validation.

4. Literature Review

The research combines work on programmable governance, evidence retrieval and document understanding. Each area contributes a different part of the proposed system.

Source	Relevance to AURA-3	Limit or research implication
Buterin (2014), Ethereum whitepaper [1]	Provides the conceptual basis for programmable state transitions and decentralized applications.	Smart contracts can enforce coded conditions. They cannot independently establish whether an off-chain business claim is true.
World Economic Forum (2023), DAO Toolkit [2]	Frames governance design and operational considerations for DAOs.	A token vote still requires careful participation rules, responsibility and risk controls.
Lewis et al. (2020), RAG [3]	Combines retrieved information with generation for knowledge-intensive tasks.	Retrieval can supply evidence, but relevance and factual support still need evaluation.
Jaume et al. (2019), FUNSD [4]	Provides a research dataset for understanding noisy scanned forms.	Form-document results do not directly establish performance on startup pitch decks.
OpenZeppelin, governance documentation [5]	Documents modular on-chain governance mechanisms, including voting and execution.	Using a standard pattern does not replace verification of the actual deployed contracts.
IPFS, content-addressing documentation [6]	Supports identifying document bytes through content identifiers.	A content identifier checks content identity. Availability needs a separate persistence or pinning arrangement.

Research interpretation

AURA-3 studies the integration boundary between these areas. Its contribution is the prototype workflow that brings extracted startup evidence into AI assessment and DAO review. The report makes no claim that this is the first such combination or that decentralization alone guarantees fair decisions.

Primary references are listed on page 23. TitanOCR-specific implementation and benchmark details come from the supplied project README [7].

5. Research Gap and Proposed Solution

Integration gap addressed by the project

Funding workflows need a consistent path from a submitted pitch deck to an assessment that a voter can inspect. In practice, extraction, reasoning and governance are distinct responsibilities. Losing the source location during extraction makes later verification difficult, while recording a vote on-chain does not validate the AI analysis that informed it.

The project addresses this engineering gap by combining a structured document contract, retrieval-based assessment and explicit governance transitions. This is a project-specific integration objective rather than a claim that all existing platforms lack these capabilities.

Proposed solution

A founder registers a startup and uploads supporting material. TitanOCR reads the document and returns a versioned Document IR, along with Markdown and text projections. The startup evaluation workflow retrieves relevant evidence, adds available market information and verification results, and produces an investor report. DAO participants review that material when considering a proposal.

The governance layer handles voting and funding records. Off-chain services retain the detailed evaluation and extraction artefacts. The association between a proposal and its supporting document version must remain stable so that a later document edit cannot silently change the evidence for an earlier decision.

Design decision	Reason
Separate extraction from understanding	OCR transcribes content and preserves locations. The downstream evaluation interprets business meaning.
Keep the Document IR authoritative	Markdown, layout text and other outputs should describe the same underlying document representation.
Retain uncertainty	Low-confidence content remains flagged so a plausible extraction error does not become an apparently certain business fact.
Keep AI advisory	Governance decides funding. A generated score is supporting analysis rather than an automatic transfer instruction.
Separate on-chain and off-chain state	Contracts enforce governance conditions. Document handling, market analysis and human review remain off-chain.

6. System Architecture and Data Flow

Layer	Components	Input and output
User interaction	React / Next.js interface, MetaMask wallet	Startup registration, document submission, proposal review and signed governance transactions.
Application services	FastAPI backend and analysis orchestration	Startup records, ingestion jobs, evaluation requests, report generation and chatbot requests.
Document processing	TitanOCR and versioned Document IR	PDF/image input becomes text, layout, tables and source-level provenance.
Retrieval and analysis	Sentence-transformer embeddings, Qdrant, RAG and market research	Relevant document evidence supports startup analysis and investor explanations.
Document storage	IPFS and application metadata	Document content identifiers and links between startup, evaluation and proposal records.
Governance	Ethereum / Solidity DAO and treasury logic	Proposal lifecycle, token-based voting and blockchain funding records.

Logical data flow

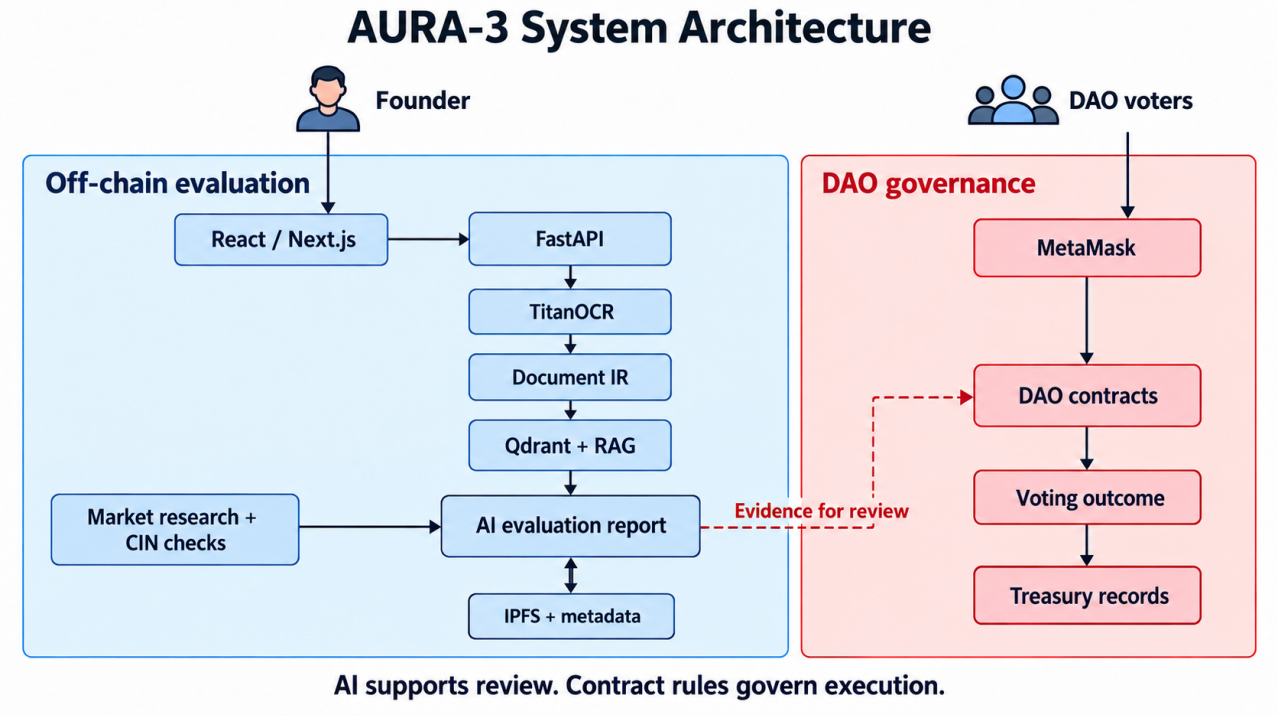
Ingestion. The backend associates an uploaded pitch deck with the startup. TitanOCR processes digital text and raster content into a common document representation. Extraction flags travel with the output.

Evaluation. Text chunks retain page and source references. Retrieval supplies context for analysis. The report distinguishes founder statements, external information and model interpretation.

Governance. Voters inspect the proposal and evaluation, then submit wallet-signed actions. The DAO records decisions and funding-related state transitions. Evidence review remains necessary for real-world milestones.

This is the logical architecture derived from the earlier report and the CA4 updates. A deployment topology, transaction trace and exact contract addresses belong in the final reproducibility package.

Figure 1. AURA-3 System Architecture



Logical architecture showing the document evaluation path and the governance workflow. AI-generated conceptual illustration based on the project architecture.

The founder submits startup details and documents through the application. FastAPI coordinates document extraction, retrieval and evaluation. TitanOCR produces the Document IR used to retain text, structure and source information. Market research and CIN checks contribute context to the evaluation report.

The governance workflow uses MetaMask for wallet access and signatures. MetaMask operates off-chain. DAO contracts record the vote outcome and funding-related state. The dashed connection represents evidence available for review, rather than authority for the AI engine to move funds.

IPFS and application metadata support the association between document content, report versions and the startup proposal. This diagram is a logical overview and does not specify the final deployment topology or contract configuration.

7. TitanOCR: Design and Implementation

The team developed TitanOCR as a standalone, general-purpose OCR and document-structure engine and utilised it in AURA-3 for pitch-deck evaluation. TitanOCR accepts PDFs and images and returns text with structure and pixel-level provenance. AURA-3 is a consumer of its output rather than the definition of the engine itself [7].

Processing path

Born-digital documents use a digital fast path. Scanned or photographed content uses detection, recognition and layout processing. The supplied implementation notes identify PP-OCRv5 detection/recognition and pinned ONNX models on the primary CPU path. Hybrid PDF pages also receive OCR for raster regions, with results merged into the same representation.

Layout and reading-order processing preserve document organisation. Table reconstruction combines geometry with an ML fallback. A versioned Document IR forms the central contract. The supplied README identifies IR version 1.8 and block types that include figures, signatures, rules, connectors and formula regions.

Provenance and uncertainty

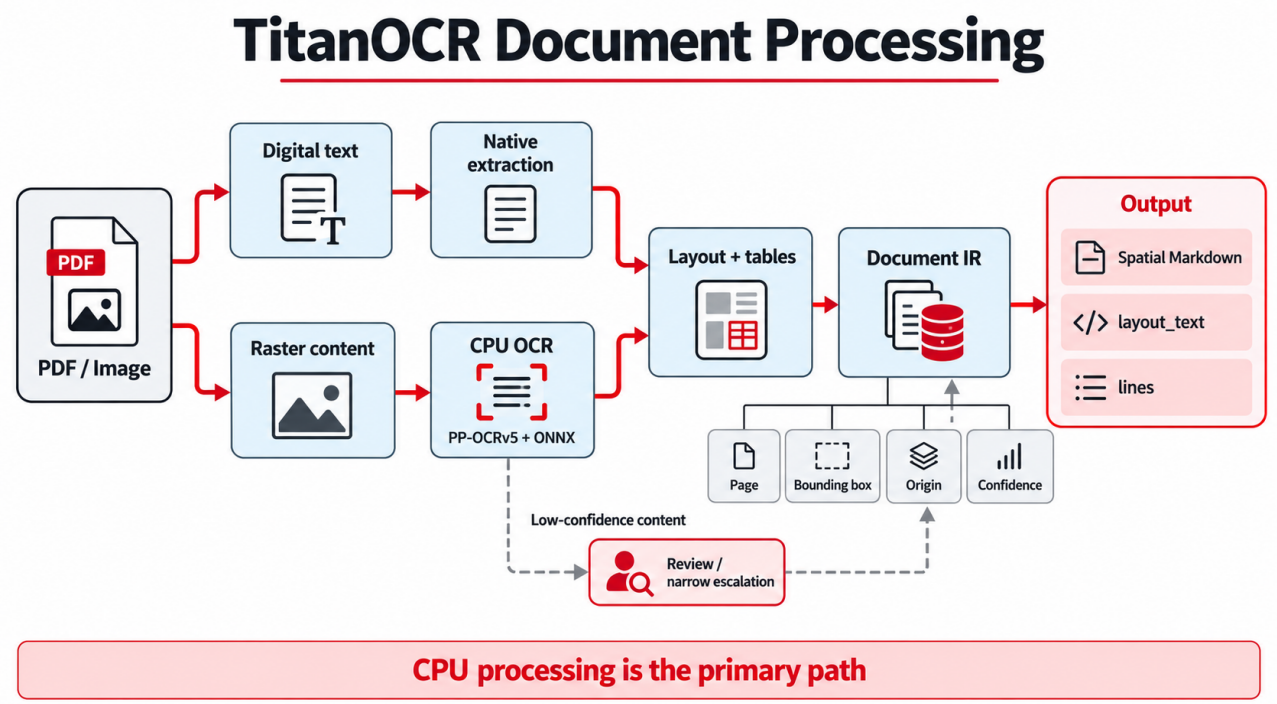
Each emitted value carries its page, bounding box, source origin and confidence. Below-threshold content is returned with a review flag or follows an escalation path. The documented design uses a VLM only as a narrow, budgeted escalation for difficult content, grounded in OCR output. Live-provider validation was blocked in the supplied status note, so this report does not treat that path as fully validated.

Execution and interfaces

The engine processes pages in a worker-process pool sized from CPU and memory resources. CPU work stays outside the asynchronous service front door. A warm engine can be reused across documents. Python and command-line interfaces expose the engine, while a JavaScript binding drives the same Python implementation.

Packaging distinction: the supplied status table says version 0.2.0 packages were built and verified but not yet published. Installation snippets in the README are therefore not evidence of public package availability.

Figure 2. TitanOCR Processing Pipeline



Digital text and raster content converge into one Document IR. AI-generated conceptual illustration based on the supplied TitanOCR design.

Digital PDFs provide native text for the fast path. Raster pages or embedded raster regions require CPU detection and recognition. Layout and table processing organise the extracted material before it reaches the common document representation.

The Document IR retains page, bounding-box, origin and confidence information. Spatial Markdown, layout_text and lines expose different views of the same evidence for downstream processing. The primary path uses CPU processing.

Low-confidence content remains visible for review or narrow, controlled escalation. The optional escalation mechanism does not replace the CPU primary path, and its live-provider validation is separate from the core extraction implementation.

8. TitanOCR in Pitch-Deck Evaluation

TitanOCR has been utilised as the extraction stage for pitch-deck evaluation in AURA-3. Its purpose is to provide the AI evaluator with readable, structured content while retaining where that content appeared. This is especially relevant for financial slides, competitor matrices and traction tables.

TitanOCR output	Use in the evaluation workflow
Document IR	Authoritative representation for page structure, source regions, recognised content and review signals.
markdown / Spatial Markdown	Preserves headings, lists and tables. Spatial Markdown adds source-line and cell comments with page, boxes, confidence and position labels.
text	Supplies words in reading order for ordinary text processing.
layout_text	Keeps visually related columns on their rows. This helps review tables containing a metric, value, unit and period.
lines	Exposes per-line text, confidence, bounding box, origin and block membership for evidence checks.
review_required	Signals that extracted evidence needs review before it supports a confident evaluation.

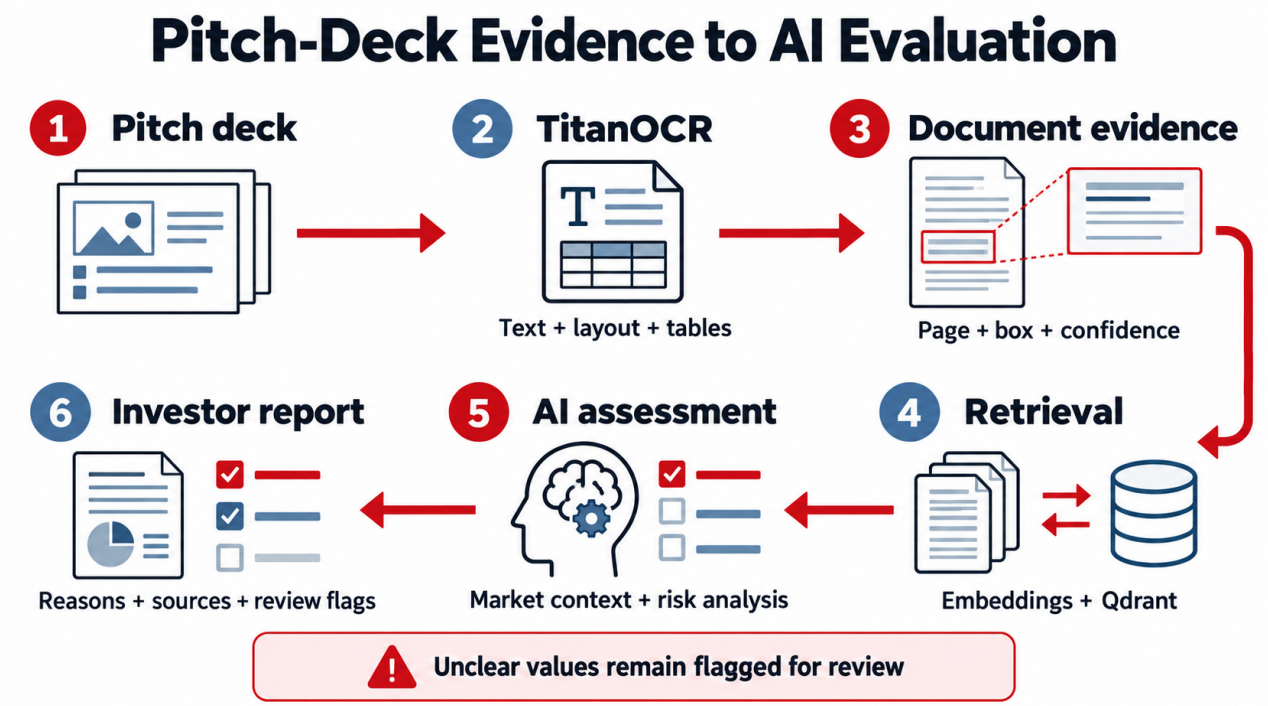
Illustrative evidence trace

Consider a pitch-deck slide that states an annual recurring revenue value beside a reporting period. The extraction layer should retain both strings and their positions. Retrieval can then present the relevant passage with the slide number. If the number is unclear, the evaluator should request review instead of inventing a value or silently accepting it.

Spatial Markdown comments must remain in the material passed to a location-aware consumer. Removing those comments can remove the coordinates and confidence that make later evidence tracing possible. Chunking should therefore preserve the link back to the source lines or IR, even when only readable text is embedded.

The example explains the integration behaviour and contains no measured startup data. TitanOCR transcribes source material. A revenue statement in a deck still requires business verification.

Figure 3. Pitch-Deck Evaluation Flow



Evidence passes from a pitch deck through extraction and retrieval into an investor report. AI-generated conceptual illustration of the AURA-3 evaluation workflow.

TitanOCR extracts the pitch-deck content with layout and table structure. Source-level evidence carries its page, location and confidence. The retrieval layer uses embeddings and Qdrant to supply relevant passages to the assessment.

The AI assessment combines retrieved document context with the available market information and risk analysis. The investor report presents reasons, source references and review flags. An unclear financial value should remain a review item instead of becoming a confident conclusion.

This flow explains the intended evidence relationship. Extraction confirms what is written in a document, while business verification establishes whether a startup claim is accurate. The illustration contains no real startup data or benchmark result.

9. AI Evaluation and Retrieval Workflow

Startup context and evidence retrieval

The backend combines startup metadata with document evidence. A sentence-transformer creates embeddings for extracted text chunks and Qdrant supports semantic retrieval. Retrieved passages supply context for the analysis and the chatbot. This continues the earlier report's RAG workflow while improving the document evidence available to it.

The analysis considers the startup's problem, product, business model, market, competition, team, traction and major risks. Market research adds external context where available. Any model-based fallback should remain distinguishable from fresh external evidence. CIN-based checks compare supplied company details with available registration information and surface mismatches for review.

Evidence handling in CA4

Stage	Information to retain
Ingest	Startup ID, document identity, extraction version and review flags.
Chunk and index	Chunk text plus page and line/region references back to the Document IR.
Retrieve	Relevant passages, their source locations and the scope of available evidence.
Analyse	Supporting passages, explicit assumptions, missing information and uncertainty.
Generate report	Readable assessment with evidence references and review items.
Answer questions	Retrieved context consistent with the selected startup and document version.

Interpretation boundary

The workflow produces decision support. Risk and return-related outputs are estimates or qualitative assessments unless independently validated. A fluent explanation, a high retrieval similarity or a confident OCR line does not prove the commercial accuracy of a founder's claim.

The earlier report describes multi-stage analysis, external research, CIN verification, report charts and a chatbot. This CA4 report does not assume a specific LLM model, embedding dimension or prompt configuration because the supplied material does not establish them.

10. DAO Implementation Completed

DAO implementation has been completed as part of the CA4 progress. It implements the governance component of AURA-3, complementing the off-chain evaluation engine. The earlier project design identifies Ethereum, Solidity, an ERC-20 governance token, Hardhat, MetaMask and Alchemy as the blockchain stack.

Governance workflow

Function	Role in AURA-3
Proposal submission	Links a startup funding request with the proposal details and supporting evaluation.
Token-based voting	Enables eligible token holders to express a governance decision under the configured voting rules.
Proposal lifecycle	Tracks the transition from a submitted request through voting and its outcome.
Treasury and funding records	Connects approved governance actions with funding-related contract state and on-chain records.
Milestone release logic	Provides the intended staged funding control. Final validation must cover milestone approval and repeated-release prevention.
Wallet and application integration	Lets participants review proposal information in the application and sign governance actions through their wallet.

Boundary between analysis and execution

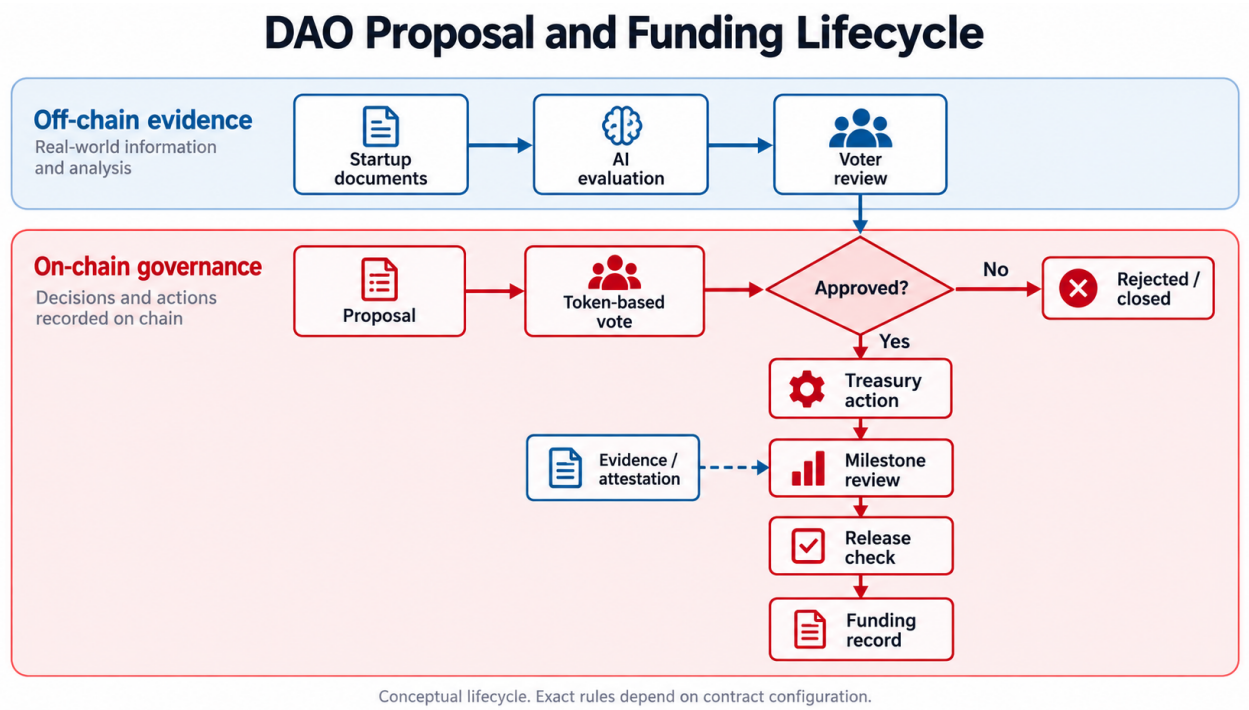
The AI engine supplies an assessment for voters. It should not independently authorize a treasury transfer. Real-world milestone completion is an off-chain fact and requires a defined evidence review or attestation process before a contract can act on it. The final validation package should identify the exact mechanism implemented.

Completion and validation

The completion statement concerns implementation progress. Contract addresses, network name, code revision, voting configuration, execution receipts and the latest test results were not included in the CA4 materials. Those artefacts remain necessary to demonstrate the complete workflow and confirm that all intended controls behave as specified.

No independent audit, mainnet readiness, specific quorum value or gas saving is claimed in this CA4 update.

Figure 4. DAO Funding Lifecycle



Conceptual lifecycle joining off-chain review with governance and funding controls. AI-generated illustration based on the project scope.

Startup documents and AI evaluation support voter review. Token holders then participate in the proposal workflow according to the configured voting rules. An unsuccessful proposal closes without an approved treasury action.

An approved outcome proceeds through the contract-defined treasury process. Milestone-based release requires evidence review or attestation and a release check. The final implementation must record and enforce the exact conditions that authorise each funding action.

The diagram summarises the project lifecycle rather than proving a deployed transaction path. The AI engine supplies supporting analysis. Wallet signatures, contract state, authorization and milestone controls govern execution.

11. Requirement Analysis and Technology Stack

Layer	Technology or capability	Purpose
Frontend	React / Next.js	Startup forms, evaluation reports, proposal review and governance interface.
API	FastAPI	Registration, ingestion, analysis, PDF reports and chatbot endpoints.
AI workflow	LangGraph-family orchestration as documented in the earlier report	Multi-stage analysis and retrieval coordination. Exact runtime version needs a build record.
Retrieval	Sentence-transformer embeddings and Qdrant	Semantic indexing and context retrieval.
Extraction	TitanOCR, ONNX models, PP-OCRv5	CPU document processing with layout, tables and source provenance.
Blockchain	Ethereum, Solidity, ERC-20, Hardhat	Governance contracts, token voting and development workflow.
Wallet / RPC	MetaMask and Alchemy	User signatures and blockchain connectivity.
Storage	IPFS and backend metadata	Content identity and application associations.
Review	CIN checks and contract-analysis workflow	Company mismatch flags and governance security review.

Functional requirements

The prototype must support startup registration, pitch-deck processing, evidence retrieval, startup evaluation, report generation, proposal review and voting. It must retain the relationship between a startup, its documents, its evaluation and its governance proposal.

Non-functional requirements

Document processing must preserve source references and expose uncertainty. The API should remain responsive during CPU-intensive OCR. Governance actions must enforce authorization and state-transition rules. Evaluation runs should record enough configuration to reproduce their results. Storage design must consider both content identity and document availability.

12. Testing and Debugging

The following matrix identifies the validation required for the CA4 additions. Implementation completion is recorded separately from test evidence. Fresh run logs and transaction receipts were not supplied, so the table gives acceptance checks rather than invented pass counts.

Area	Acceptance check	Evidence to retain
Digital pitch decks	Selectable text preserves its content, page and position.	Source file, extracted IR and line comparison.
Scanned / hybrid decks	Raster text is recognised and merged without losing digital content.	Paired page render and extraction output.
Tables and units	Numbers remain associated with the correct metric, unit and period.	Human-labelled cell/line truth and mismatch report.
Low-confidence content	Uncertain text exposes a review flag or controlled escalation.	Confidence trace, routing output and review decision.
RAG and chatbot	Answers refer to the correct startup and supporting source.	Retrieved passages, response and human support labels.
DAO voting	Eligibility, voting window and vote accounting follow configured rules.	Unit/integration logs and transaction receipts.
Treasury workflow	Rejected or invalid actions cannot release funds. A completed release cannot repeat.	Contract tests, balances and emitted events.
End-to-end association	The voted proposal refers to the intended evaluation version.	Startup, document, report and proposal identifiers.

Debugging priorities

Investigate lost table headers, swapped columns, small rotated labels and OCR errors in numeric strings. For governance, focus on invalid transitions, repeated actions, wallet rejection and differences between displayed application state and confirmed contract state. Any low-confidence business-critical value should remain visible as a review item.

13. Research Evidence and Evaluation Limits

TitanOCR's supplied README reports development evidence on research datasets and throughput measurements. These results are useful for identifying progress and failure cases, but they are not an end-to-end accuracy claim for AURA-3 pitch-deck evaluation [7].

Reported experiment	Result in supplied README	Interpretation
ICDAR 2013 table stratum	125 digital PDFs and 24,333 labelled cells. Cell accuracy at (row, col) improved from 17.3% to 24.0% to 39.7%.	Table reconstruction improved, but substantial errors remain. These are table-stratum measurements.
FUNSD scanned forms, initial stratum	199 forms. Labelled lines found: 98.4%. Bbox-matched line CER: 10.2%. Review required: 88% of pages.	High line coverage coexists with recognition errors and a high review rate under provisional thresholds.
Later recognition experiment	Reported FUNSD line CER improved from 8.08% to 6.45%.	A later experiment with its own baseline. It should not be merged into the earlier run as one comparable series.
Throughput on a 16 GB M5 laptop	About 1.5-2 minutes for a 40-page scanned document. About 15 seconds for a 40-page digital PDF.	Hardware-specific repository measurements, not a service SLA or pitch-deck benchmark.
Release evaluation	Harness includes 14 gates. Real human-labelled release golden set not yet available in the supplied status.	Overall release verdict remains incomplete. General accuracy claims are unsupported.

AURA-3 evaluation required next

Build a human-labelled set of digital, scanned and hybrid pitch decks. Measure character error rate, financial-field accuracy, table association accuracy, source-location correctness and the rate of claims supported by evidence. For governance, record valid and invalid lifecycle transitions and treasury invariants.

Earlier report figures such as 84.2% AI-panel agreement, 60.5% voter participation and 1.2-second IPFS retrieval are not repeated as CA4 results because their run definitions and supporting logs were not supplied.

14. Results

This section reports a complete run of the AURA-3 prototype carried out on 26 October 2026. Both services were started locally and the application was used through a browser in the order a founder and then an investor would use it. Each figure is a screenshot captured during that session. Each number in the tables comes from the application's own logs, its stored evaluation record, the exported report or a direct read of the Sepolia contract. The run is a single demonstration case rather than a benchmark: it shows what the implemented system did with one pitch deck, and where it stopped.

Test setup

Component	Configuration in this run
Frontend	Next.js 16.2 (React 19) development server on localhost:3000. Repository revision a85e7c1 with uncommitted changes.
Backend	FastAPI service, API version 3.0.0, on localhost:8010. Repository revision 0b53d5b with uncommitted changes.
Language model	azure/deepSeek-V4-Flash through an OpenAI-compatible gateway; embeddings from text-embedding-004.
Document extraction	Docling, with pypdf and pdfplumber as fallbacks. TitanOCR is not called by this build.
Retrieval and search	Per-startup vector index of 1,000-character chunks; web search through the DuckDuckGo text, news and Lite endpoints.
Company registry	Sandbox registry. No MCA provider key was configured, so the backend returned its built-in placeholder record.
Blockchain	Ethereum Sepolia. VentureDAO contract 0xC25A8B17eCf3FaB710E74188E4136220E713Fd89, state read at block 11,998,010.
Input	KrishiMitra_AI_Pitch_Deck.pdf (11 slides, 16 KB) and a founder profile taken from the deck: Agritech, Seed, ask USD 1,440,000, valuation USD 7,200,000, CIN U62099MH2024PTC410567.
Client	Google Chrome 156 (headless), 1440 × 900 viewport, scripted with Playwright so that each step was captured as it happened.

End-to-end evaluation run

The founder form was submitted once. The table lists what the backend recorded, timed from the moment the startup record was created.

Step	What the system recorded	Time
Registration	Startup record created. CIN checked against the sandbox registry: ACTIVE, KRISHIMITRA AI PRIVATE LIMITED, no compliance flags.	0.0 s
Ingestion	Docling extracted 3,486 characters from the deck; 5 chunks were embedded and indexed.	6.6 s
Research, round 1	The analyst agent planned searches. The grader rejected the evidence: no market size or CAGR with currency, unit, year and source URL.	17.8 s
Research, round 2	The query was rewritten and searched again. The grader rejected it for the same reason.	33.9 s
Research, round 3	The second rewrite was searched. The rewrite limit of two was reached, so the workflow moved on to scoring.	44.1 s
Scoring	Seven category scores and the market-sizing step completed.	55.0 s
Report writing	Two writer agents produced the 11 report sections in parallel.	77.6 s
PDF export	Investor PDF built from the stored research snapshot: 9 pages, 103 KB.	About 1 s after request

Registration to finished report took 77.6 seconds. The two rejected research rounds account for 37 seconds of that time. This is the grader working as designed: it did not pass forward research that lacked a sourced market figure, and the bounded loop still let the run finish. During the same window several requests to one search provider failed with a network error; the search layer fell back to other DuckDuckGo endpoints and the run continued. Figure 6 shows the founder's view of this phase, including the grader's message streamed live from the backend.

Evaluation output

The evaluation scored KrishiMitra AI 7.2 out of 10 with a Buy label. The category scores are the model's analytical judgments, not measured quantities.

Category	Score / 10	Category	Score / 10
Product	8.0	Business model	8.5
Market	7.5	Moat	6.5
Team	7.0	Execution risk (10 = low risk)	6.0
Traction	8.0	**Overall**	**7.2 (Buy)**

The report contains all 11 sections and lists 35 web sources, each with its retrieval time. Six items were declared as data gaps (Figure 9). TAM, SAM, SOM and CAGR were shown as unavailable, and the

market-trend and revenue charts were left empty, because none of the retrieved search excerpts carried a usable figure with value, unit and period. The charts therefore did not plot the deck's own figures as market evidence.

The written report is less strict than the charts. Its Market Opportunity section restates the deck's TAM of USD 82.3 million, with the deck's MarketsandMarkets citation, without saying that the figure came from the pitch deck rather than from retrieved research. A reader of the PDF sees an attributed market size in the text and an empty market-sizing chart beside it. Figures taken from the company's own material should be labelled as company claims in the narrative as well.

The MCA verified badge in Figures 7 and 8 reflects the sandbox registry, not the Ministry of Corporate Affairs. The sandbox returns a fixed record whose directors (Arjun Malhotra, Priya Nair) are placeholders and do not match the founding team. The consistency checks compare company name and status but not directors, so no flag was raised. The badge demonstrates the verification flow, not the company's registration.

The same deck had been evaluated four times before on this installation, with the same model and slightly different profiles. The five overall scores span 2.0 points.

Run date	Profile used	Overall score
24 Sep 2026	Agritech, Seed, ask USD 1,000,000	5.2 (Hold)
24 Sep 2026	Agritech, Seed, ask USD 100,000	6.8 (Buy)
24 Sep 2026	Fintech, Seed, ask USD 100,000	6.8 (Buy)
24 Sep 2026	Agritech, Series A, ask USD 1,000,000	6.2 (Buy)
26 Oct 2026	Agritech, Seed, ask USD 1,440,000 (this run)	7.2 (Buy)

Because the profiles differed, this is not a controlled repeatability test. It does indicate that the score should not yet be read at one-decimal precision. A repeated run on a fixed profile is needed before scores are shown to voters as a ranking.

Investor-facing assistants

Two conversational paths were tested after the evaluation had finished.

Assistant	Question asked	Observed behaviour	Time
Venture Analyst (per-startup chat)	What are the three biggest risks for KrishiMitra AI, and what evidence supports them?	Answered from the stored evaluation and the deck: execution risk in expanding beyond Maharashtra, churn sensitivity (2.4% monthly churn, 4.1-month payback), and competitive and funding pressure (Figure 12).	4.4 s

Assistant	Question asked	Observed behaviour	Time
FinScope advisory	Explain TAM, SAM and SOM for an agritech startup, with a simple example.	Intent classified as education and routed to the Financial Educator subagent. The routing decision and its reason were streamed to the page (Figure 13).	3.9 s

Both answers addressed the question asked. The FinScope example uses round illustrative numbers, such as 500 million farms worldwide, without sources. That is reasonable in an explanation, but these figures are illustrations rather than evidence.

Governance state on Sepolia

The investor dashboard reads proposals directly from the VentureDAO contract (Figure 14). To check what the interface showed, all 18 proposals were also read from the contract at block 11,998,010. They were created between 24 March and 15 September 2026.

Outcome, as the application derives it	Proposals	Detail
Executed	3	Proposals 6, 10 and 11. Largest 'for' total: 0.01201 GOV.
Passed, not executed	2	Proposal 7 (0.00001 GOV for) and proposal 18 (0.001 GOV for).
Ended without a majority	13	12 with no votes cast; proposal 5 with 0.00001 GOV against.

The full lifecycle has taken place on the deployed contract: creation, a timed vote, a majority outcome and execution. These records say little about governance behaviour, because the proposals are development tests, funding requests range from 0.00000001 to 500 test ETH, and turnout was close to zero. No new proposal, vote or execution was submitted in this run, because no wallet was connected.

The run exposed two display problems. Vote totals are rounded to two decimals, so proposal 18 shows 0.00 for and 0.00 against while labelled Passed (Figure 15); the contract holds 0.001 GOV for. The ETH/USD price route returned HTTP 503 throughout the session, so USD values were shown as '$---'. ETH amounts were unaffected.

Results against the CA4 demonstration checklist

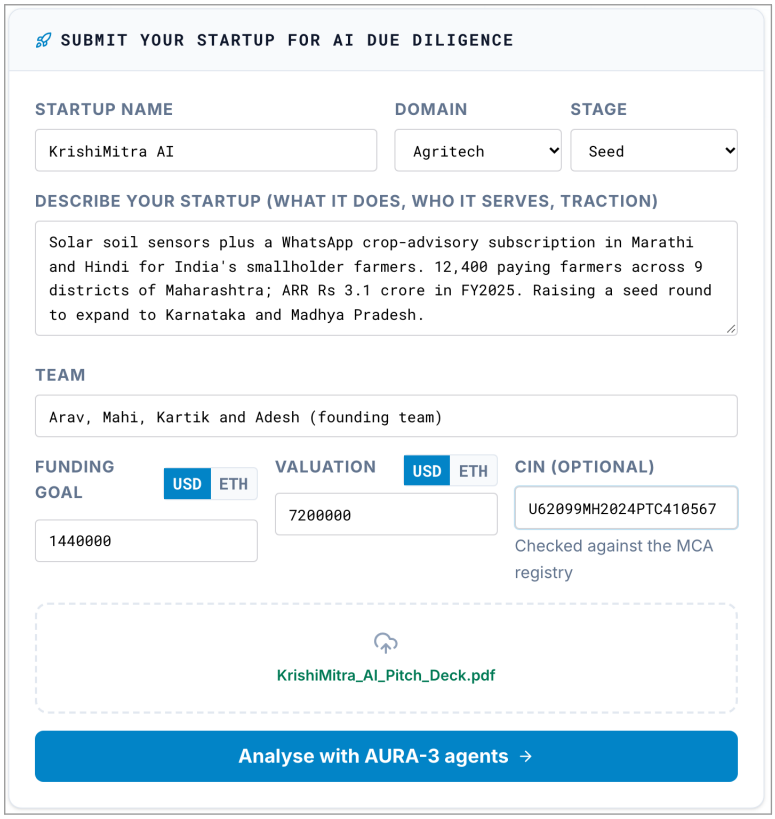
Checklist item	Status after this run	Evidence
Startup intake	Shown	Figure 5 and the stored startup record.
TitanOCR processing	Not shown	This build extracts with Docling; no Document IR or per-line confidence was produced.

Checklist item	Status after this run	Evidence
Evidence verification	Not shown	Chunks keep the source file and chunk number, not page or bounding box.
AI evaluation	Shown	Figures 6 to 11: 35 sources, six declared data gaps, PDF export.
Chatbot	Shown	Figures 12 and 13.
DAO interaction	Partly shown	Proposals read from the contract (Figures 14 and 15); no wallet transaction in this run.
Governance outcome	Partly shown	Proposal states read on-chain; no transaction receipt captured.
Funding control	Partly shown	Three executed proposals exist; invalid-release scenarios were not exercised.
Reproducibility	Partly shown	Model, contract, block and revisions recorded above; both repositories had uncommitted changes.

In summary, the off-chain evaluation path ran end to end on a real pitch deck in under 80 seconds, its charts declined to plot market figures that the search evidence did not support, and both assistants answered in under five seconds. The open items match those already named in CA4: the build that was run does not yet call TitanOCR, the registry check runs against a sandbox, score repeatability is untested, and on-chain governance activity is at test scale.

Figures 5 to 15 follow. Scores and narrative in the screenshots are AI output and are reproduced as generated.

Figure 5. Founder portal: startup submission



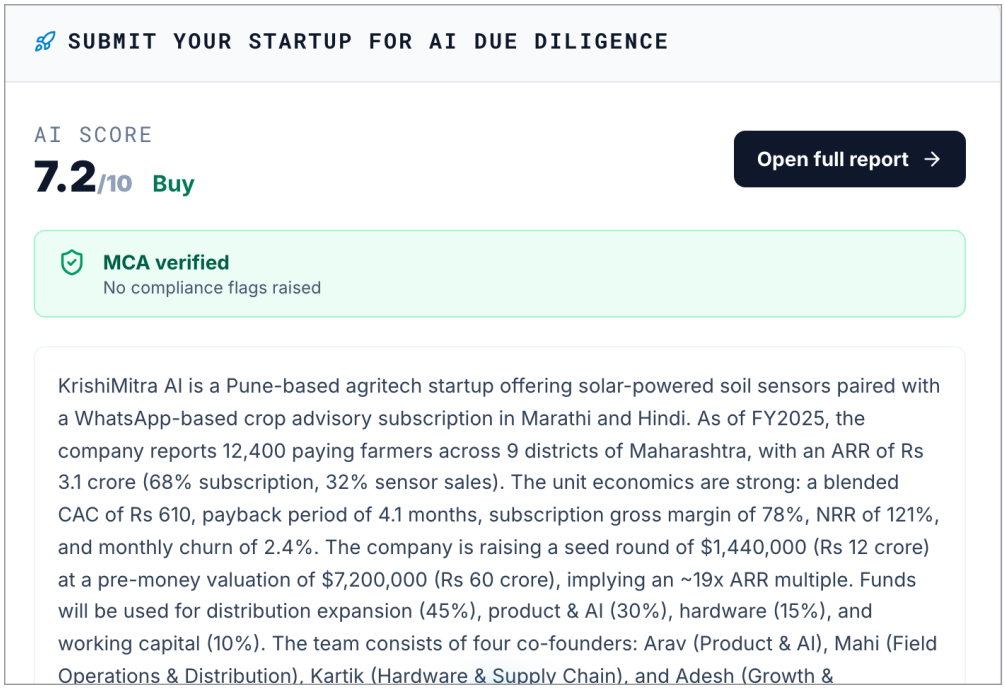
The founder enters the profile taken from the pitch deck, the CIN for the registry check and the funding terms, then attaches KrishiMitra_AI_Pitch_Deck.pdf. 'Analyse with AURA-3 agents' starts registration, ingestion and evaluation in one step.

Figure 6. Multi-agent evaluation in progress



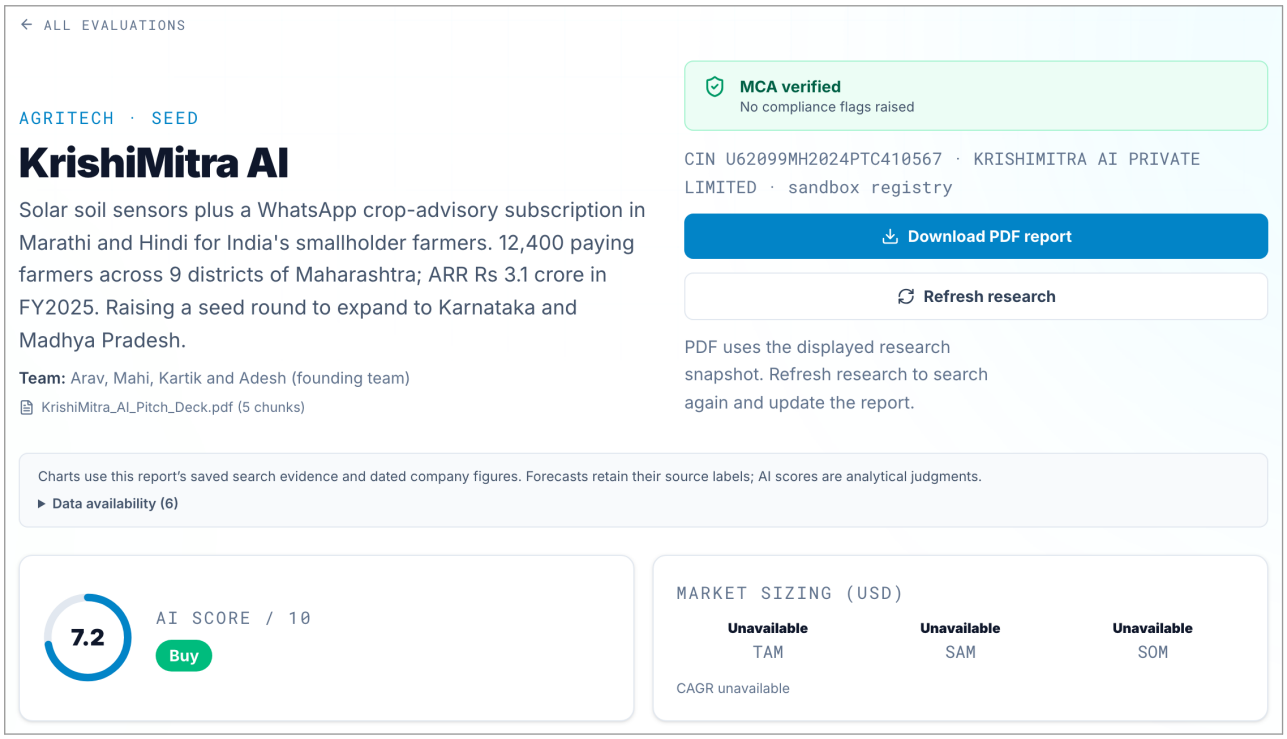
Captured about 47 s after submission. Three LangGraph stages are complete and scoring is running. The log panel streams the grader's decision over Socket.IO: research was judged insufficient because no market size or CAGR came with a source URL, which triggered a query rewrite.

Figure 7. Evaluation complete in the founder portal



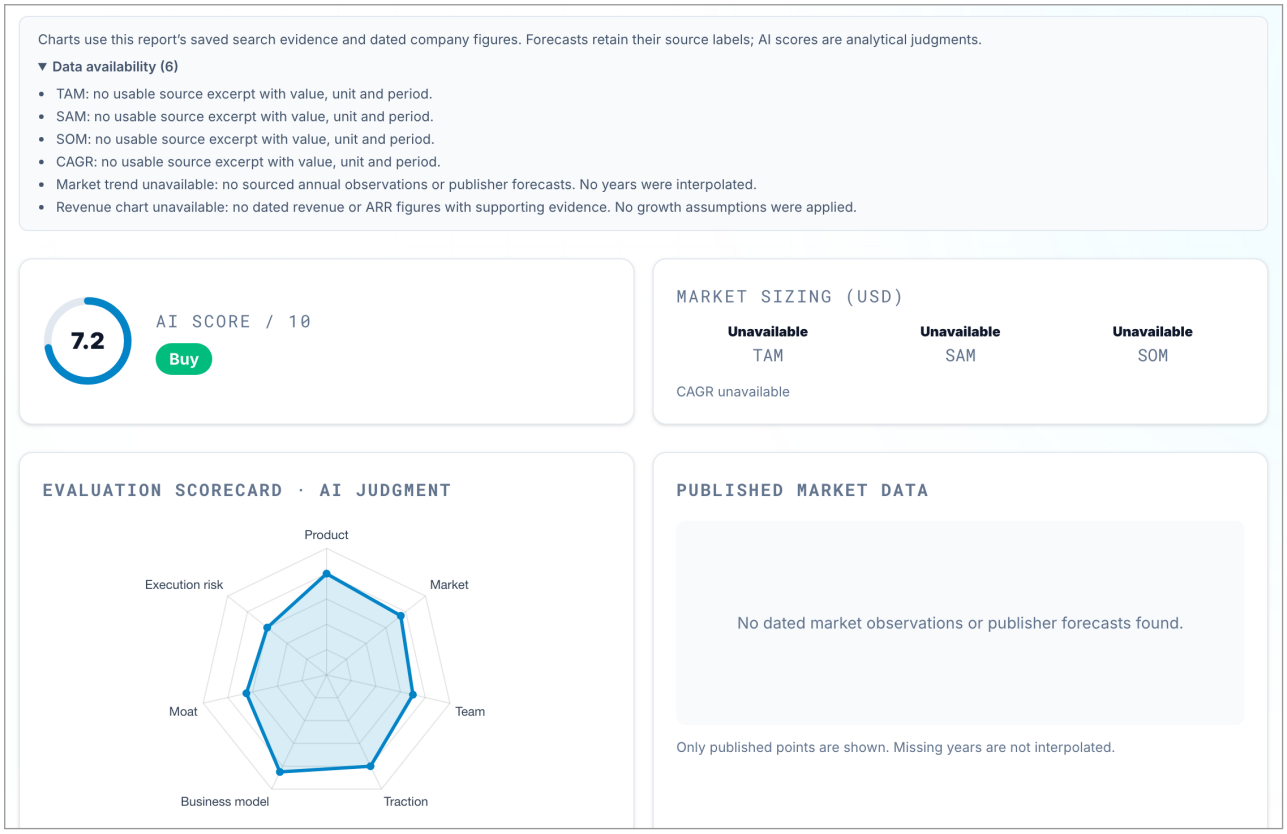
The founder sees the score, the registry badge and the executive summary as soon as the run finishes, with a link to the full report. The badge comes from the sandbox registry.

Figure 8. Evaluation page: profile, registry result and score



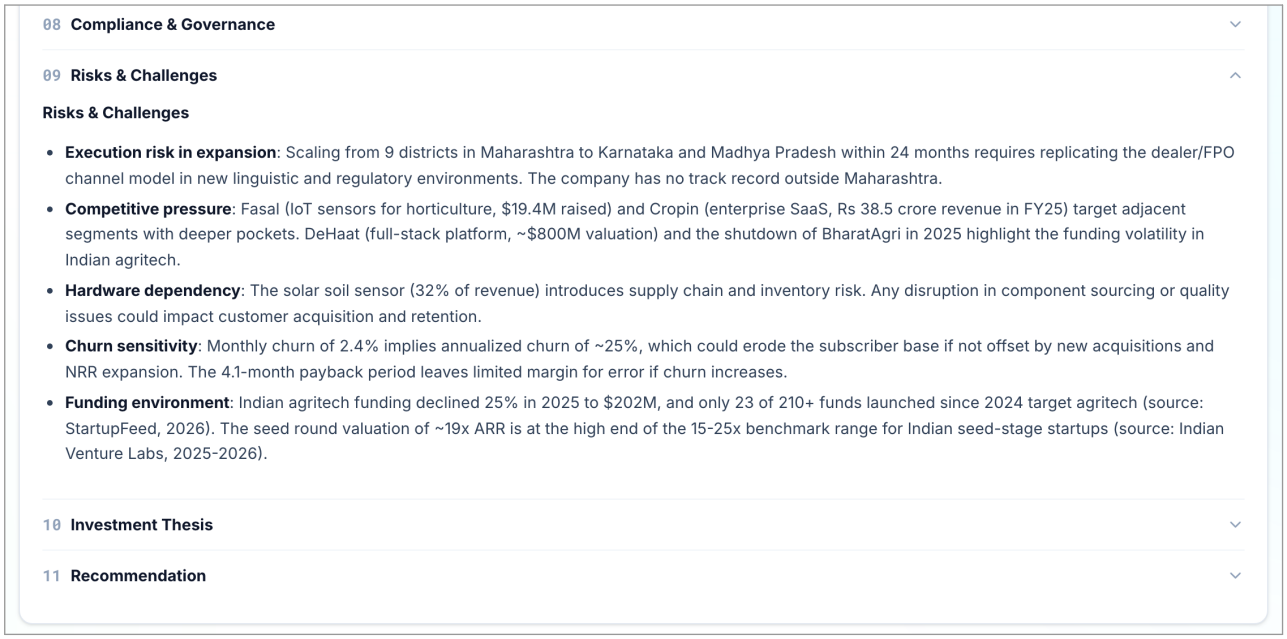
The CIN line is labelled 'sandbox registry'. PDF export and research refresh are on the right. The market-sizing card reports TAM, SAM and SOM as unavailable rather than estimating them.

Figure 9. Declared data gaps and evaluation scorecard



The six data gaps are disclosed above the charts. The radar chart plots the seven category scores and is labelled as AI judgment.

Figure 10. Risks section of the 11-section investor report



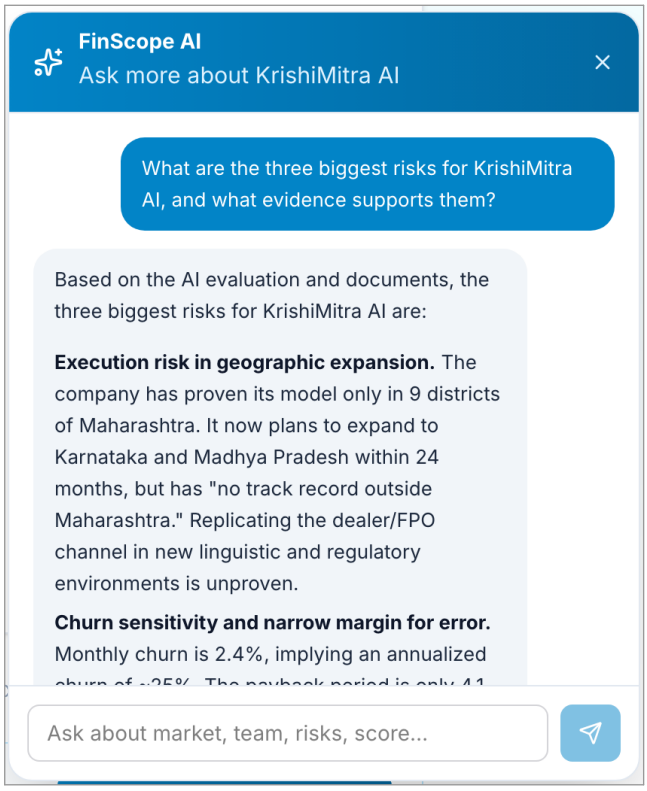
Risks cite named competitors and funding figures from the retrieved sources, such as Fasal, Cropin and DeHaat, and derive churn and valuation concerns from the deck's own numbers.

Figure 11. Exported investor PDF, pages 1 to 3



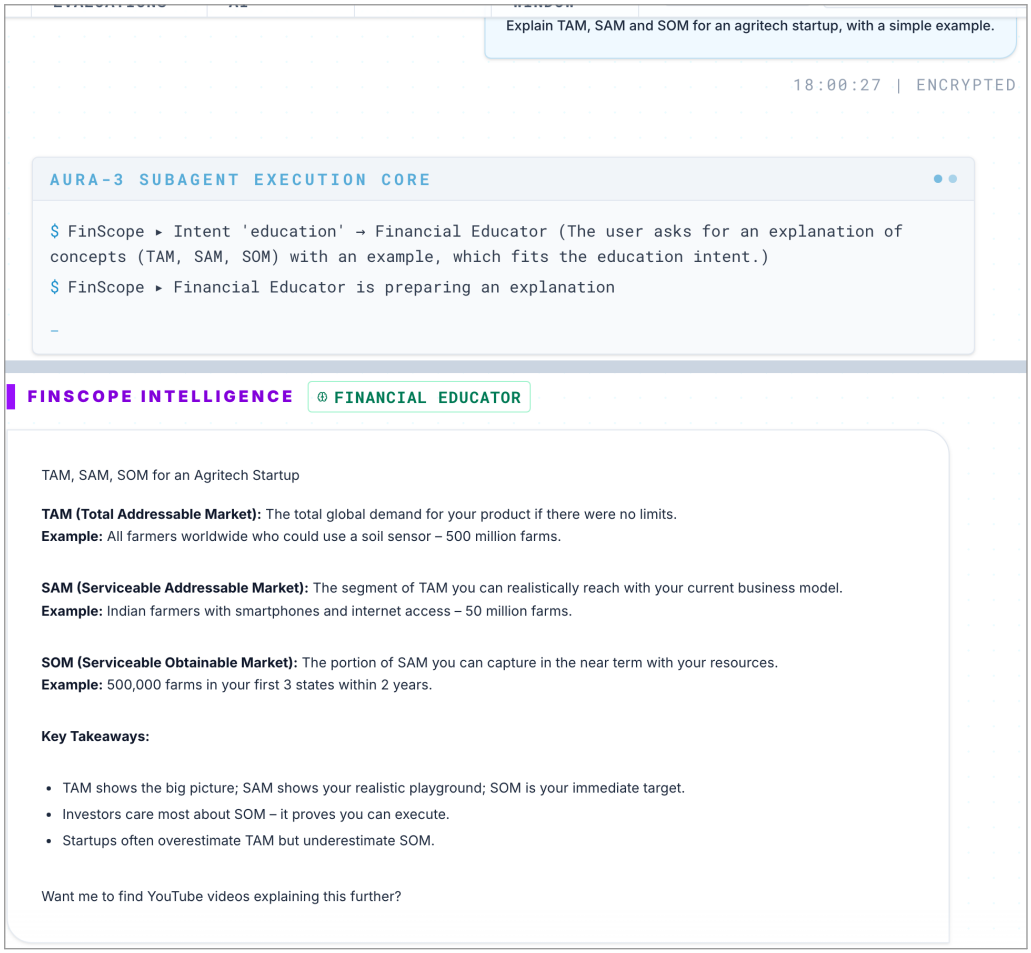
Upper halves of the first three pages of the 9-page PDF: profile and registry table, data-gap disclosure, and the scorecard chart.

Figure 12. Venture Analyst chat



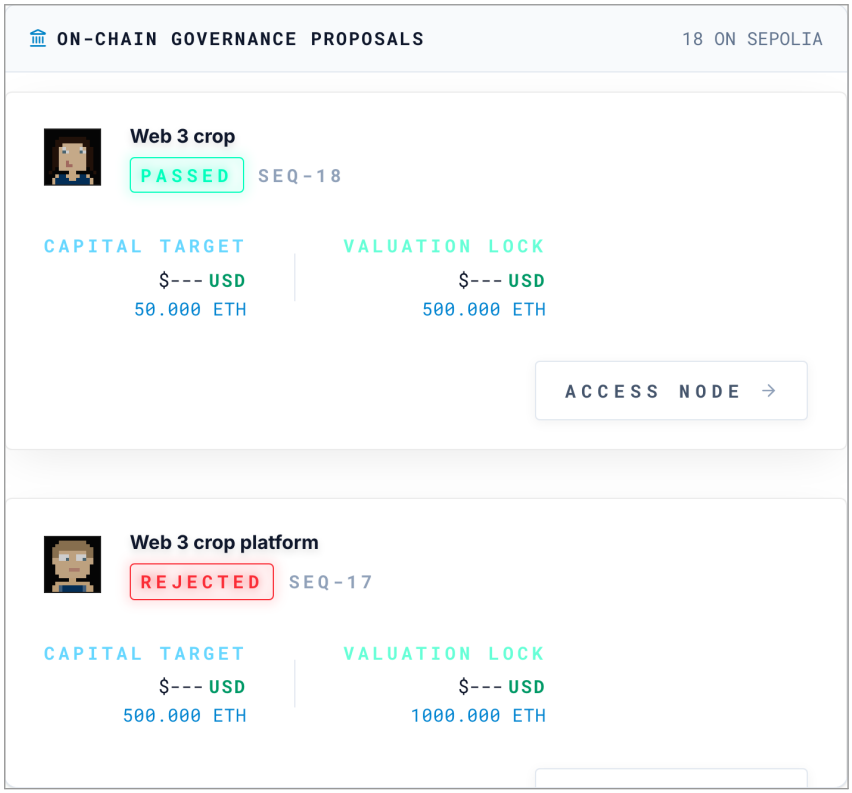
The investor's question and the start of the answer. The assistant draws on the stored evaluation and the deck; it answered in 4.4 s.

Figure 13. FinScope routing and answer



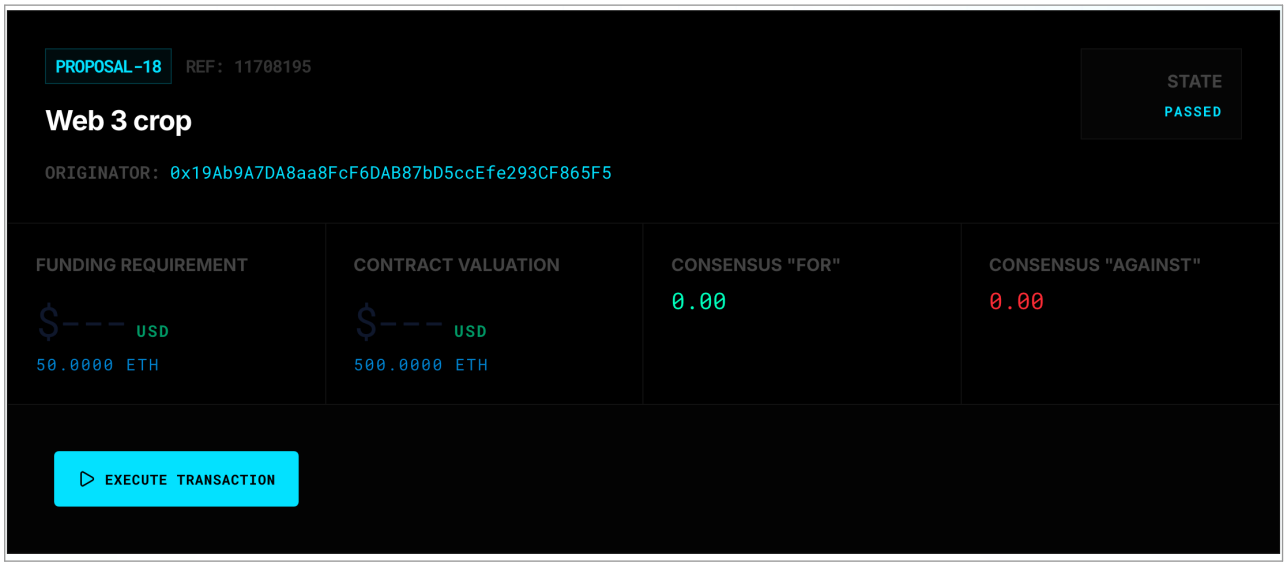
Top: the subagent execution log shows the intent classifier routing the question to the Financial Educator, with its reason. Bottom: the answer, tagged with the subagent that produced it.

Figure 14. Investor dashboard: proposals read from Sepolia



The dashboard lists 18 proposals from the VentureDAO contract with their derived state. USD values show '$---' because the ETH/USD price route returned HTTP 503 during the session.

Figure 15. Proposal 18 detail



State 'Passed' with tallies displayed as 0.00. The contract holds 0.001 GOV for and 0 against, so the two-decimal display hides the deciding vote.

15. Project Plan and CA4 Progress

Workstream	Earlier baseline	CA4 progress / next action
Research and requirements	Literature survey, problem definition and architecture.	Research framing retained. Evaluation questions now include document provenance and uncertainty.
Application and AI	Frontend, FastAPI, Qdrant/RAG, analysis, reports and chatbot described in earlier report.	Workflow extended to utilise TitanOCR for pitch-deck extraction.
TitanOCR development	Not documented as a named engine in the earlier report.	Engine developed and utilised. Structured output and spatial provenance now support the document-processing design.
DAO governance	DAO and treasury design with blockchain integration described.	DAO implementation completed. Demonstration and reproducibility evidence to be consolidated.
Integration validation	Preliminary testing narrative in the earlier report.	End-to-end source/version links, voting and release scenarios remain the next validation focus.
Final documentation	Monthly report and presentation baseline.	CA4 report and presentation updated with DAO implementation and TitanOCR research detail.

Next milestones

Integration evidence. Capture one complete startup journey from pitch-deck upload to evaluation, proposal review, voting and funding-related contract state.

Research evaluation. Finalise the pitch-deck dataset and labels, record the OCR/LLM configuration and compare ordinary extraction with provenance-preserving processing.

Governance verification. Record contract deployment details, role and voting configuration, negative-path test results and milestone-release behaviour.

Final submission. Consolidate screenshots, references, run artefacts and required institutional appendices. The order is a work plan rather than a claim that these milestones have already been completed.

16. Project Outcomes and Limitations

Outcomes achieved by CA4

Completed DAO implementation. The governance implementation is now part of the reported project progress. It complements startup evaluation with proposal, voting and funding-related workflow capabilities.

TitanOCR developed and utilised. The project now uses its own standalone document-processing engine for pitch-deck evaluation. The extraction contract exposes text, layout and evidence locations instead of relying only on a flattened text string.

Research-aligned document handling. Spatial provenance, explicit review flags and a distinction between transcription and interpretation give the team a clearer basis for evaluating evidence quality. These capabilities support auditability without proving that an extracted business claim is true.

Updated submission material. The CA4 documentation brings the AI, document-processing and DAO components into one account of the implemented prototype and remaining validation.

Limitations

The supplied CA4 materials contain no labelled pitch-deck benchmark, current contract deployment manifest or complete end-to-end run logs. General OCR accuracy, investment quality, bias reduction, security and voting-participation effects therefore remain unproven.

Small fonts, charts, complex tables, handwriting and poor photographs can still cause extraction failures. Recognising labels around a chart does not establish the meaning or correctness of the chart. Spatial coordinates and OCR confidence are evidence attributes, not calibrated probabilities that a business statement is correct.

Token-weighted governance may concentrate influence. Off-chain evidence and market information can be incomplete or outdated. IPFS content identifiers do not guarantee permanent availability. These constraints should inform evaluation and future design, without being represented as resolved by the current prototype.

17. Conclusion and Next Steps

CA4 advances AURA-3 from the earlier architecture and evaluation pipeline toward a more complete academic prototype. The DAO implementation has been completed, and the team has developed and utilised TitanOCR for pitch-deck evaluation. Together, these additions address the document-evidence and governance parts of the funding workflow.

TitanOCR contributes a CPU-based primary extraction path, a versioned Document IR and output formats that preserve source information. The AI workflow consumes that material to support analysis and retrieval. The DAO provides the governance process through which participants consider a proposal. Each layer has a distinct responsibility and a separate validation requirement.

The research contribution at this stage is the implemented integration and the explicit treatment of evidence provenance. Stronger conclusions require a controlled pitch-deck evaluation, contract-level verification and an end-to-end demonstration with reproducible artefacts. Reported TitanOCR research numbers should remain tied to their original datasets and runs.

Immediate next steps

1. Freeze a demonstrable build and record the application, OCR and contract revisions.

2. Capture source documents, extraction output, retrieved evidence and the generated evaluation for a representative startup case.

3. Demonstrate the DAO lifecycle with both accepted and rejected actions, including the treasury and milestone checks actually implemented.

4. Complete a human-labelled pitch-deck evaluation and report uncertainty, review rate and failure categories alongside average quality.

5. Consolidate final documentation and institution-generated similarity or AI reports only when authentic reports become available.

CA4 completion statement: DAO implementation completed. TitanOCR developed and utilised for pitch-deck evaluation. Final empirical validation and supporting artefacts remain the next project milestone.

18. References

[1] Buterin, V. (2014). *A next-generation smart contract and decentralized application platform*. Ethereum. https://ethereum.org/whitepaper/

[2] World Economic Forum. (2023). *Decentralized Autonomous Organization Toolkit*. https://www.weforum.org/publications/decentralized-autonomous-organization-toolkit/

[3] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.-t., Rocktaschel, T., Riedel, S., & Kiela, D. (2020). *Retrieval-augmented generation for knowledge-intensive NLP tasks*. Advances in Neural Information Processing Systems, 33. https://arxiv.org/abs/2005.11401

[4] Jaume, G., Ekenel, H. K., & Thiran, J.-P. (2019). *FUNSD: A dataset for form understanding in noisy scanned documents*. ICDAR Workshops. https://arxiv.org/abs/1905.13538

[5] OpenZeppelin. (n.d.). *How to set up on-chain governance*. Contracts 5.x documentation. Retrieved October 15, 2026, from https://docs.openzeppelin.com/contracts/5.x/governance

[6] IPFS. (n.d.). *Content identifiers (CIDs)*. IPFS documentation. Retrieved October 15, 2026, from https://docs.ipfs.tech/concepts/content-addressing/

[7] AURA-3 project team. (2026). *TitanOCR: Project README and implementation status* [Unpublished project documentation supplied for CA4]. Source attachment: Pasted text(20261015-054918).txt. Metrics and status are attributed to the supplied text, not independently rerun.

[8] Jain, A., Saxena, A., Mehta, K., & Laddha, M. (2026). *AURA-3: AI-powered Decentralized Autonomous Organization (DAO) for startup funding. Monthly progress report - August 2026* [Unpublished project report]. Symbiosis Institute of Technology, Pune.

Source handling

The institutional details and earlier implementation baseline follow the supplied monthly report. CA4 completion updates follow the project team's request. TitanOCR capabilities, version status and development measurements follow the supplied README. Primary research and documentation references support the design rationale.

Appendix: CA4 Demonstration Checklist

This checklist specifies the artefacts to include with a final demonstration. It is a preparation aid and does not state that the artefacts have already been collected.

Demonstration step	Artefact or observation
Startup intake	Startup metadata and the exact uploaded pitch deck.
TitanOCR processing	Document IR, Markdown / Spatial Markdown, per-line confidence and any review flag.
Evidence verification	One extracted financial or traction statement traced back to its page and bounding box.
AI evaluation	Retrieved passages, market source references, assumptions and generated investor report.
Chatbot	A response supported by evidence from the selected startup document.
DAO interaction	Proposal identifier, wallet interaction and configured voting behaviour.
Governance outcome	Confirmed transaction receipt and proposal state.
Funding control	Treasury state and valid/invalid release scenarios for the implemented rules.
Reproducibility	Code revisions, model/configuration versions, network and contract addresses, test logs.

Institutional appendices

The earlier template lists a similarity report and an AI plagiarism report. These must be authentic outputs from the applicable institutional process. They are not generated or simulated in this CA4 report.

Project identification

Project ID: 24
Department of Artificial Intelligence & Machine Learning
Symbiosis Institute of Technology, Pune
Guide: Dr. Wasim Khan
Co-guide: Dr. Shivali Wagle